UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

6657998344

PHYSICS **5054/02**

Paper 2 Theory **May/June 2008**

1 hour 45 minutes

Candidates answer on the Question Paper.

Additional Materials: Answer Booklet/Paper.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions.
Write your answers in the spaces provided on the Question Paper.

Section B
Answer any **two** questions.
Write your answers on the lined pages provided, and, if necessary, continue on the separate answer paper provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
Q9	
Q10	
Q11	
Total	

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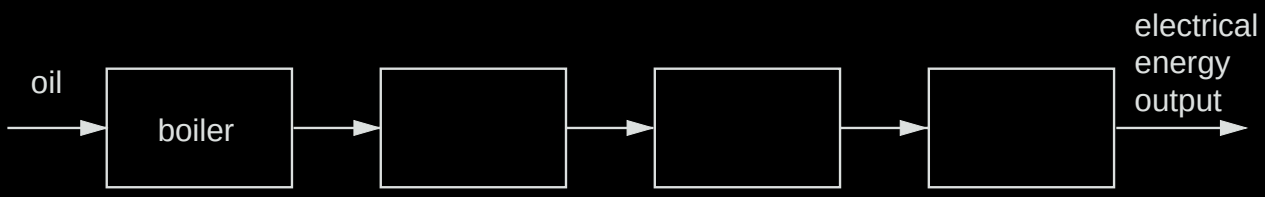


Fig. 1.1

Each of the other three boxes should contain **one** of the labels from the following list.

generator, motor, transformer, turbine, solar panel

(a) On Fig. 1.1, label the boxes using words from the list. [2]

(b) State **one** environmental problem caused by burning oil to produce electricity.

.....

..... [1]

(c) Oil is a non-renewable energy source.

(i) State why oil is described as a non-renewable energy source.

.....

.....

..... [1]

(ii) State **one** renewable energy source.

.....

..... [1]

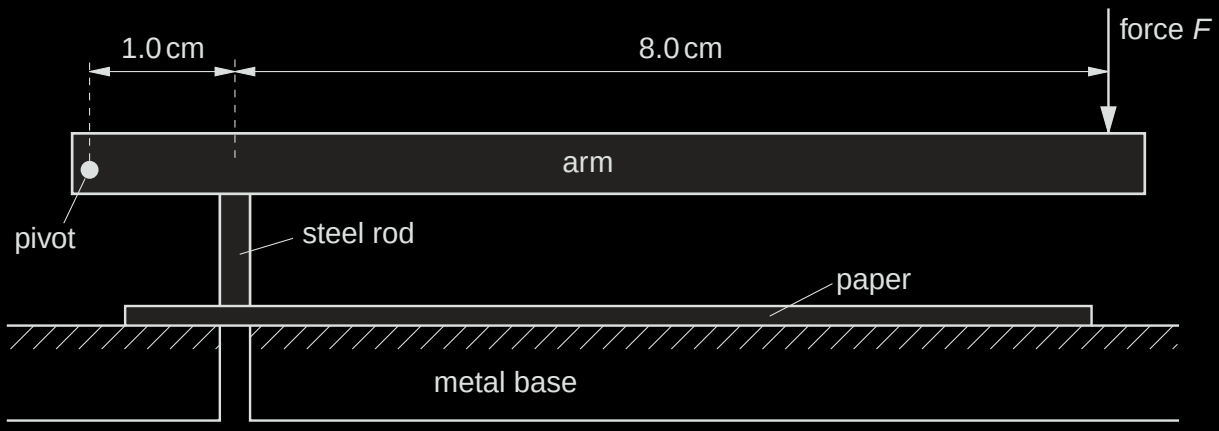


Fig. 2.1

(a) Just before the hole is made, the force upwards on the steel rod is 7.2 N.

Calculate the value of *F*. Use the distances marked on Fig. 2.1.

F = ... [2]

(b) The steel rod pushes the paper into the hole in the metal base with a force of 7.2 N. The end of the steel rod has an area of $2.8 \times 10^{-5}\,m^2$.

Calculate the pressure exerted on the paper.
State the formula that you use in your calculation.

pressure = ... [2]

(c) The force downwards on the paper due to the rod and the force upwards on the rod due to the paper are related by Newton's third law. State Newton's third law.

.....

.....

..... [1]





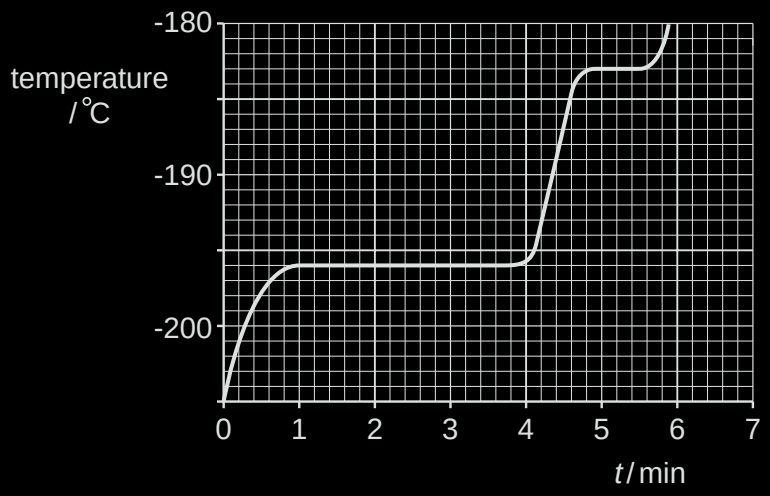


Fig. 4.1

(a) Describe the two changes of state that occur and state when they occur.

.....

.....

.....

..... [3]

(b) The liquid air contains 200 g of liquid oxygen and 800 g of liquid nitrogen.
The specific heat capacity of liquid oxygen is 1.7 J/(g °C) and the specific heat capacity of liquid nitrogen is 2.0 J/(g °C).
Calculate the thermal energy needed to warm the liquid from –205 °C to −196 °C.
State the formula that you use in your calculation.

energy = ... [4]

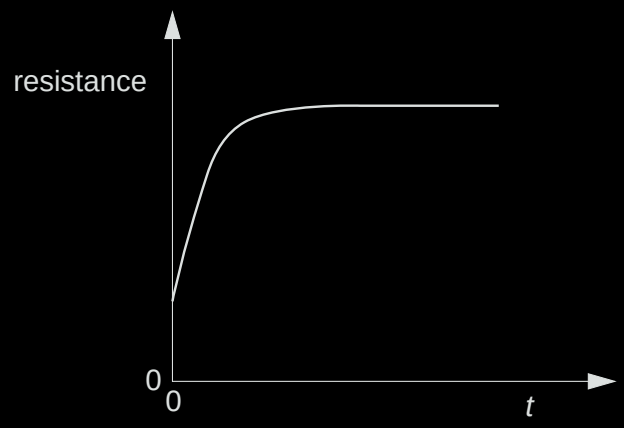


Fig. 6.1

Describe how the current in the lamp varies with *t*.

.....

.....

.....

.....

..... [2]

(c) A second filament lamp has higher resistance.

State **two** differences between the two filaments that may cause the change in resistance.

1 The filament in the second lamp is .....

.....

2 The filament in the second lamp is .....

..... [2]

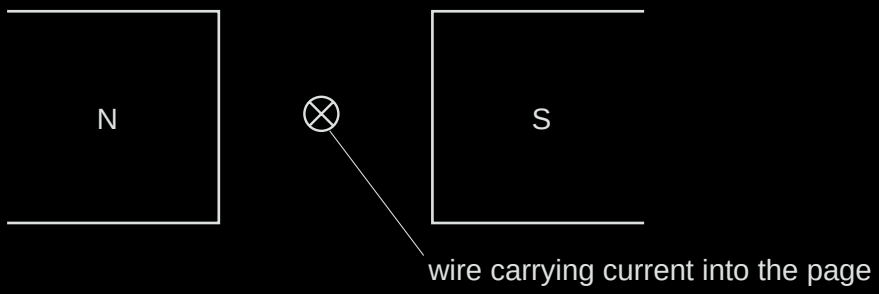


Fig. 7.1

(i) State the direction of the magnetic field between the poles of the magnet.

..... [1]

(ii) On Fig. 7.1, draw an arrow to show the direction of the force acting on the wire. [1]

(b) Fig. 7.2 shows two wires.
Each wire carries a current into the page.

Fig. 7.2

(i) On Fig. 7.2, draw the magnetic field due to the currents in the wires. [3]

(ii) There is a force on each wire due to the current in the other wire.

On Fig. 7.2, draw an arrow on each wire to show these forces. [1]

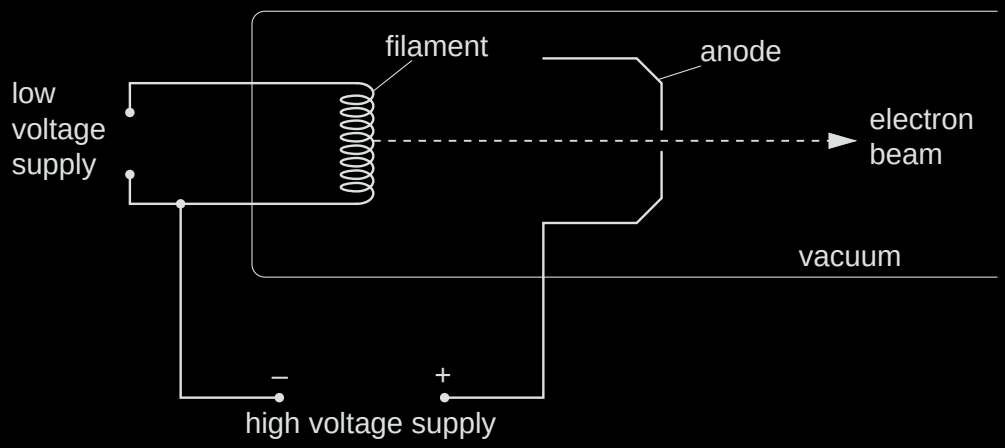


Fig. 8.1

(a) State why the filament emits electrons.

..... [1]

(b) State

(i) how the electrons are made to move at high speed,

.....

.....

..... [1]

(ii) why a vacuum is needed inside the oscilloscope.

.....

.....

..... [1]

(c) In each second, 8.0×10^{14} electrons pass through the hole in the anode.
The charge on an electron is 1.6×10^{-19} C. Calculate the current in the electron beam.

current = ... [2]

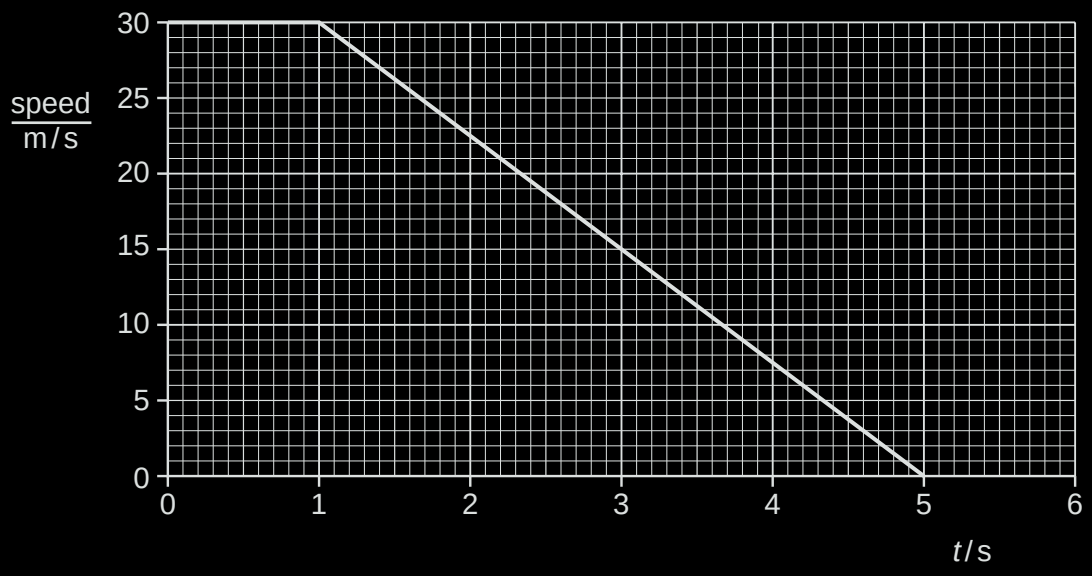


Fig. 9.1

Determine

(i) the thinking distance, [1]

(ii) the braking distance, [2]

(iii) the deceleration of the car during braking, [2]

(iv) the force provided by the brakes. [2]

(c) Using ideas about friction and deceleration, state and explain how the braking distance is affected by

(i) using new tyres rather than badly worn tyres, [2]

(ii) the car skidding on a wet road, [2]

(iii) the car carrying a heavy load of passengers. [2]

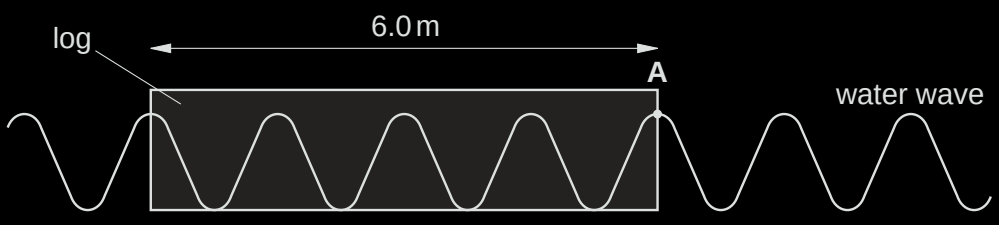


Fig. 10.1

The log is 6.0 m long and 5 complete waves take 10 seconds to pass point **A**.

Determine

(i) the wavelength of the water waves, [1]

(ii) the frequency of the water waves, [2]

(iii) the speed of the water waves. [2]

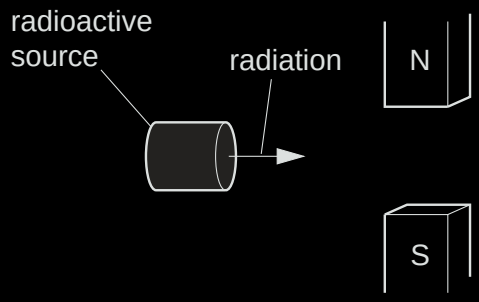


Fig. 11.1

The radioactive source emits alpha-particles, beta-particles and gamma-rays. For each type of radiation, describe the path taken by the radiation.
You may draw a diagram to help your explanation. [3]

(c) Fig. 11.2 represents three different nuclei.

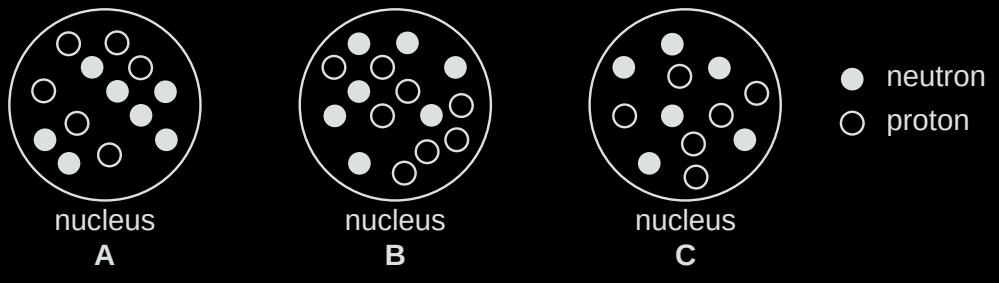


Fig. 11.2

State and explain which nuclei are from different isotopes of the same element. [3]